MAT357
Foundations of Real Analysis

Class Lecture Notes

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

III Measure Theory

8 Basics

riemann integral =(. "im kind of biased in saying this, but im also correct, so". fire.

Definition 8.1

for an int'vl $I \subseteq \mathbb{R}$ w endpts $a < b$, define $\ell(I) = b - a$. for $A \subseteq \mathbb{R}$, its **outer measure** $m^*(A)$ is:

$$m^*(A) := \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : A \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \text{ open bdd int'vls} \right\}$$

note $m^*(A) \in [0, \infty]$, i.e. $\mathbb{R}_{\geq 0}$ but can also be infty.

Proposition 8.2

i) if $A \subseteq B$, $m^*(A) \leq m^*(B)$

ii) if E ctbl, $m^*(E) = 0$

iii) if I int'vl, then $m^*(I) = \ell(I)$

Proof.

i) any cvr of B cvrs A.

ii) if $E = \{x_i\}$, let $I_k = (x_k - \varepsilon/2^{k+1}, x_k + \varepsilon/2^{k+1})$. then:

$$m^*(E) \leq \sum \ell(I_k) = \sum \varepsilon/2^k = \varepsilon$$

iii) sps $J = [a, b]$ (renamed). sps I_k open int'vls cvring J. J cpt, so take finitely

many; also assume no $I_j \subseteq I_k$. if $I_k = (a_k, b_k)$ and $\{a_i\}, \{b_i\}$ increasing, then:

$$\sum_{j=1}^{k} \ell(I_j) \;=\; \sum_{j=1}^{k} b_j - a_j \;=\; b_k - a_1 + \sum b_j - a_{j+1} \;>\; b_k - a_1 \;\geq\; b - a$$

so $\forall$ cvrs $\{I_j\}$, $\sum \ell(I_j) \geq b - a$ implies $m^*(J) \geq b - a$. OTOH, if $J_\varepsilon = (a - \varepsilon, b + \varepsilon)$, then $m^*(J) \geq \ell(J_\varepsilon) \to b - a$. [i love aboos of notation]

now, if $J = (a, b)$, let $I^- = [a + \varepsilon, b - \varepsilon]$ and $I^+ = [a - \varepsilon, b + \varepsilon]$. clearly, $I^- \subseteq J \subseteq I^+$, so $m^*(I^-) \leq m^*(J) \leq m^*(I^+)$, and it follows.

■

Lemma 8.3

if $E \subseteq \mathbb{R}$ and $y \in \mathbb{R}$, then $m^*(E + y) = m^*(E)$.

translate ur cvrs.

Lemma 8.4

if $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $m^*(E) \leq \sum m^*(E_k)$.

Proof.

for all k, let $\left\{I_j^{(k)}\right\}$ be an int'vl cvr of E_k s.t.:

$$m^*(E_k) \;\leq\; \sum_{j=1}^{\infty} \ell(I_j^{(k)}) \;\leq\; m^*(E_k) + \frac{\varepsilon}{2k}$$

since $\left\{I_j^{(k)}\right\}_{j,k}$ an int'vl cvr of E, then:

$$m^*(E) \;\leq\; \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\ell(I_j^{(k)}) \;\leq\; \sum_k m^*(E_k) + \frac{\varepsilon}{2k} \;\leq\; \sum_k m^*(E_k) \;+\; \varepsilon$$

■

Definition 8.5

a set $E \subseteq \mathbb{R}$ is **measurable** if $\forall A \subseteq \mathbb{R}$:

$$m^*(A) \;=\; m^*(A \cap E) + m^*(A \cap E^c)$$

we denote by $\mathcal{E}$ the set of measurable sets.

note by the union property, we always have $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$. thus, to show E measurable, we only need to show:

$$m^*(A) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

note also, non-measurable sets exist; that is, $\exists A, B$ disjoint s.t.:

$$m^*(A \cup B) \;\lneq\; m^*(A) + m^*(B)$$

Lemma 8.6

if $m^*(E) = 0$, then $E \in \mathcal{E}$.

Proof.

note $m^*(A \cap E) + m^*(A \cap E^c) \;\leq\; m^*(E) + m^*(A) \;=\; m^*(A)$. ■

Lemma 8.7

if $E_1, E_2 \in \mathcal{E}$, then $E_1 \cup E_2 \in \mathcal{E}$.

Proof.

sps $A \subseteq \mathbb{R}$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

since $A \cap (E_1 \cup E_2) \;=\; (A \cap E_1) \cup (A \cap E_1^c \cap E_2)$. ■

Definition 8.8

a coll'n $\mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$ is an **algebra** if:

- $\mathbb{R} \in \mathcal{A}$
- $E \in \mathcal{A} \implies E^c \in \mathcal{A}$
- $E_1 \cup E_2 \in \mathcal{A} \implies E_1 \cup E_2 \in \mathcal{A}$

Corollary 8.9

$\mathcal{E}$ is an alg

Proposition 8.10

if $C_1, C_2 \in \mathcal{E}$ disj, then $m^*(C_1 \cup C_2) = m^*(C_1) + m^*(C_2)$.

Proof.

note $A \cap E_1 \;=\; A \cap E_1 \cap E_2 + A \cap E_1 \cap E_2^c$, so:

$$\begin{aligned} m^*(A \cap (E_1 \cup E_2)) &= m^*(A \cap (E_1 \cup E_2) \cap E_2) + m^*(A \cap (E_1 \cup E_2) \cap E_2^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_2) \end{aligned}$$

■

Proposition 8.11

$\mathcal{E}$ closed under ctbl unions.

Proof.

let $A_j \in \mathcal{E}$. denote:

$$E_1 \;=\; A_1 \qquad E_j \;=\; A_j \setminus \bigcup_{k=1}^{j-1} A_k$$

then E_j disj and $E = \bigcup E_j$. let $F_n = \bigcup_{j=1}^n E_j$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap F_n) + m^*(A \cap F_n^c) \\ &\geq m^*(A \cap F_n) + m^*(A \cap E^c) \\ &= \sum_{j=1}^n m^*(A \cap E_j) + m^*(A \cap E^c) \end{aligned}$$

taking $n \to \infty$:

$$m^*(A) \;\geq\; \sum_{j=1}^\infty m^*(A \cap E_j) + m^*(A \cap E^c) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

■

Definition 8.12

a σ-algebra is an alg closed under ctbl unions.

thus, $\mathcal{E}$ is a σ-alg. note also that $\mathcal{E}$ closed under ctbl intersections by taking complements.

Definition 8.13

we call $\mathcal{E}$ the set of **Lebesgue measurable** sets, and denote $m := m^*|_{\mathcal{E}}$ as the **Lebesgue measure**.

Lemma 8.14

$(a, \infty) \in \mathcal{E}$.

Proof.

let $A \subseteq \mathbb{R}$; first sps $a \notin A$.

sps $\{I_j\}$ cvrs A. let $J_k = I_k \cap (-\infty, a), K_k = I_k \cap (a, \infty)$. note:

$$\sum_j \ell(I_j) \;=\; \sum_j \ell(J_j) + \sum_j \ell(K_j)$$

we also have:

$$A \cap (-\infty, a) \;\subseteq\; \bigcup_j J_j \qquad A \cap (a, \infty) \;\subseteq\; \bigcup_j K_j$$

therefore, we have that:

$$m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \;\leq\; \sum_j \ell(J_k) + \sum_j \ell(K_j) \;=\; \sum_j \ell(I_j)$$

and it follows that $m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \leq m^*(A)$. ■

Lec 16 - Mar 10 (Week 9)

missed

Lec 17 - Mar 17 (Week 10)

midterm last thurs

Theorem 8.15

i) if $A_n \subseteq A_{n+1} \in \mathcal{E}$ then $m\left(\bigcup A_n\right) = \lim_n m(A_n)$

ii) if $m(B_1) < \infty$ and $B_n \supseteq B_{n+1}$ then $m\left(\bigcap B_n\right) = \lim_n m(B_n)$

Proof.

i) let $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} = A_n \cap (A_{n-1})^c \in \mathcal{E}$. then they are disj, so:

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = m\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} m(C_n)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} m(C_k) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{n} C_k\right) = \lim_{n\to\infty} m(A_n)$$

ii) let $A_n = B_1 \setminus B_n$. since B_n is decreasing, A_n is increasing, so

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} m(A_n)$$

note:

$$m(A_n) = m(B_1 \setminus B_n) = m(B_1) - m(B_n)$$

OTOH:

$$\begin{aligned} m\left(\bigcup_{n=1}^{\infty} A_n\right) &= m\left(\bigcup_{n=1}^{\infty} B_1 \setminus B_n\right) \\ &= m\left(B_1 \setminus \bigcap_{n=1}^{\infty} B_n\right) \\ &= m(B_1) - m\left(\bigcap_{n=1}^{\infty} B_n\right) \\ &= \lim_{n\to\infty} m(B_1) - m(B_n) \end{aligned}$$

■

recall the defn of a σ-alg. given a coll'n of subsets of $\mathbb{R}$ the σ-alg *generated by* A is the intersection of all σ-algs containing A, denoted $\sigma(A)$.

Definition 8.16

a **measure space** $(\Omega, \mathcal{F})$ is a set Ω and a σ-alg $\mathcal{F}$ on Ω. given two measure spaces $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$, we say $f : \Omega_1 \longrightarrow \Omega_2$ is **measurable** if $\forall E \in \mathcal{F}_2$, we have that $f^{-1}(E) \in \mathcal{F}_1$.

note the similarity to (topological) continuity, not that the similarity is particularly important though.

Lemma 8.17

if $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$ are measure spaces s.t. $\mathcal{F}_2 = \sigma(A)$ for some coll'n $A \subseteq \mathcal{P}(\Omega_2)$, then $f : \Omega_1 \longrightarrow \Omega_2$ is measurable if:

$$f^{-1}(E) \in \mathcal{F}_1$$

for all $E \in A$.

Proof.

suffices to check that

$$S := \{E : f^{-1}(E) \in \mathcal{F}_1\}$$

is a σ-alg containing A. indeed, its a σ-alg essentially bc $\mathcal{F}_1$ is.

$$E_n \in S \implies f^{-1}\left(\bigcup_{n=1}^{\infty} E_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(E_n) \in \mathcal{F}_1 \implies \bigcup_{n=1}^{\infty} E_n \in S$$

other axioms are similar. ■

so why do we care abt this lemma?

Definition 8.18

the **Borel $\boldsymbol{\sigma}$-alg** is given as $\mathcal{B}(\mathbb{R}) := \sigma(\text{open subsets of } \mathbb{R})$.

fact: $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{E} \subsetneq 2^{\mathbb{R}}$. the inclusions being strict is tedious and unnecessary to show, and thus skipped.

Definition 8.19

we say a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is **Lebesgue measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{E}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. we say it is **Borel measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{B}(\mathbb{R})) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Lemma 8.20

a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is lebesgue (borel) meas if:

$$f^{-1}((c, \infty)) \in \mathcal{E} \qquad (\text{or } \mathcal{B}(\mathbb{R}))$$

for any $c \in \mathbb{Q}$.

Proof.

suffices to check $\{\text{open sets}\} \subseteq \sigma(\{(c, \infty)\}) =: \mathcal{F}$ for $c \in \mathbb{Q}$.

i) check $(a, b) \in \mathcal{F}$. indeed, $(a, b) = (a, \infty) \setminus [b, \infty)$, but $[b, \infty) = \bigcap (q, \infty)$ for $q \in \mathbb{Q} \cap (-\infty, b)$.

ii) any open set in $\mathbb{R}$ is a ctbl union of open intvls w/ rational endpts.

■

Corollary 8.21

any cts or monotonic $f : \mathbb{R} \longrightarrow \mathbb{R}$ is borel measurable.

Proposition 8.22

sps $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ are lebesgue (borel) meas. the following are as well:

i) $|f|, \alpha f + \beta$ for real α, β

ii) $(f)_+, (f)_-$ where $(x)_*$ is the $+$ve or $-$ve parts of x

Proof.

these are compositions of f w/ a cts fn, hence borel meas. easy to see that the composition of a BM fn w/ a LM fn is LM.

■

Lec 18 - Mar 19 (Week 10)

Proposition 8.23

in addition to the above, the following are also lebesgue (borel) meas.:

iii) $f + g, f - g, fg$

iv) $\max(f, g), \min(f, g)$

Proof.

iii) we show $f+g$ is lebesgue. suffices to check $\{x : f(x)+g(x) > c\} \in \mathcal{E}\ \forall c \in \mathbb{R}$. we claim that:

$$\{x : f(x)+g(x) > c\} \in \mathcal{E} \ = \ \bigcup_{q \in \mathbb{Q}} \{x : f(x) > c-q\} \cap \{x : g(x) > q\}$$

enough to check $(\subseteq)$. indeed, sps x is s.t. $f(x)+g(x) > c$. then $\exists q \in \mathbb{Q}$ with $g(x) > q > c - f(x)$, with $f(x) > c-q$ and $g(x) > q$.

$f-g$ is similar, so we show fg. note:

$$fg \ = \ \frac{1}{4}\left((f+g)^2 - (f-g)^2\right)$$

wts if f meas, then f^2 is, but this is just a composition of meas. fns.

iv) follows from above, as:

$$\max(f,g) \ = \ \frac{|f-g|+(f+g)}{2} \qquad \min(f,g) \ = \ \frac{(f+g)-|f-g|}{2}$$

■

it is sometimes useful to consider fns which take values $\pm\infty$.

Definition 8.24

let $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ denote the extended real line. we define:

$$\mathcal{B}(\overline{\mathbb{R}}) \ := \ \left\{E \subseteq \overline{\mathbb{R}} : E = A \cup B, A \in \mathcal{B}(\mathbb{R}), B \in \{\varnothing, \{\infty\}, \{-\infty\}, \{\pm\infty\}\}\right\}$$

Exercise 8.25

i) $\mathcal{B}(\overline{\mathbb{R}})$ is a σ-alg

ii) $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is LM (BM) iff $\{x : f(x) > c\} \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall c \in \mathbb{Q}$

iii) if $f : \mathbb{R} \longrightarrow \mathbb{R}$ meas., then $\iota \circ f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas.

any meas. fn defined on the ext. reals is (unsurprisingly) also called LM/BM. the difference only rly matters when it comes to operations on fns (whats $\infty - \infty$?).

given $E \in \mathcal{E}$, $f : E \longrightarrow \overline{\mathbb{R}}$, tfae:

- f meas
- $f^{-1}(A) \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall A \in \mathcal{B}(\overline{\mathbb{R}})$
- the fn $\widetilde{f} = \chi_E f$ is meas.

equivalence of last one is an exercise.

9 Limits of measurable functions

Proposition 9.1

let $f_n : E \longrightarrow \overline{\mathbb{R}}$ a seq of ptwise incresing meas. fns for some E meas. then $\lim_{n\to\infty} f_n(x) = f(x)$ is meas.

Proof.

for all $c \in \mathbb{R}$, since f_n is ptwise increasing, we have:

$$\{x : f(x) > c\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > c\}$$

■

Theorem 9.2

sps E meas., $f_n : E \longrightarrow \overline{\mathbb{R}}$ seq of meas. fns. the following are also meas.:

i) $x \mapsto \sup_n f_n(x)$

ii) $x \mapsto \inf_n f_n(x)$

iii) $x \mapsto \limsup_n f_n(x)$

iv) $x \mapsto \liminf_n f_n(x)$

v) $x \mapsto \lim_n f_n(x)$, provided this exists $\forall x$

Proof.

i) $\forall c \in \mathbb{R}$:

$$\left\{x : \sup_n f_n(x) > c\right\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > c\}$$

ii) similar as above

iii) recall $\limsup_n f_n(x) = \lim_n \left\{\sup_{m \geq n} f_m(x)\right\}$. this is a decreasing seq of meas. fns, so by i), we're done

iv) similar as above

v) if the lim exists, its equal to both limsup and liminf

■

we know what a.e. is (uses outer meas. specifically, since we dk a priori the set is meas)

Lemma 9.3

sps $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ LM and $g : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is s.t. $f(x) = g(x)$ ae. then g meas.

Proof.

let $E = \{x : f(x) \neq g(x)\}$. for $A \in \mathcal{B}(\overline{\mathbb{R}})$, we have:

$$g^{-1}(A) \;=\; (f^{-1}(A) \cap E^c) \cup F$$

where $F = g^{-1}(A) \cap E$. then $m^*(F) \leq m^*(E) = 0$, so $F \in \mathcal{E}$. by assumption, $f^{-1}(A) \in \mathcal{E}$ and $E^c \in \mathcal{E}$, so $g^{-1}(A) \in \mathcal{E}$. ■

Corollary 9.4

sps $f_n : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas s.t. $\lim_n f_n(x)$ exists ae. let $f(x) = \lim_n f_n(x)$ when it exists, and arbitrarily otw. then f LM.

Proof.

let $E = \{x : \lim_n f_n(x) \text{ exists}\}$. note $E \in \mathcal{E}$ since $m^*(E^c) = 0$. defn:

$$\widetilde{f_n}(x) \;=\; \begin{cases} f_n(x) & x \in E \\ 0 & \text{otw} \end{cases}$$

then $\widetilde{f}(x) = \lim_n \widetilde{f_n}(x)$ exists $\forall \mathbb{R}$. note:

i) $\widetilde{f_n}(x)$ LM since $\widetilde{f_n} = \chi_E f_n$

ii) $\widetilde{f}$ LM since its ptwise lim of LM fns

thus $f = \widetilde{f}$ a.e., so by lemma, f is LM. ■

Exercise 9.5

show E defn'd as above is always meas., even if the lim doesnt necessarily exist ae.

hint: lim exists iff limsup = liminf.

10 Simple functions

throughout this section, $E \in \mathcal{E}$ unless stated otw.

Definition 10.1

an LM $\phi : E \longrightarrow \mathbb{R}$ is **simple** if $|\mathrm{im}(\phi)| < \infty$. equivalently:

$$\phi = \sum_{k=1}^{n} \alpha_k \chi_{E_k}$$

where $\alpha_k \in \mathbb{R}, E_k \in \mathcal{E}$.

Lemma 10.2

sps $f : E \longrightarrow \mathbb{R}$ bdd LM, $\varepsilon > 0$. then $\exists \phi_\varepsilon, \psi_e$ simple s.t. $\phi_\varepsilon \leq f \leq \psi_\varepsilon$, and $|\psi_e - \phi_e| < \varepsilon$ for all x.

Proof.

let $M > |f(x)|$ for all $x \in E$, and write:

$$(-M, M] = \bigcup_{j=1}^{n} (y_{j-1}, y_j]$$

s.t. $|y_j - y_{j-1}| < \varepsilon$ for some strictly increasing y_j. let $E_j = f^{-1}((y_{j-1}, y_j])$. by choice of M:

$$E = \bigsqcup_{j=1}^{n} E_j \qquad \phi_\varepsilon = \sum_{j=1}^{n} y_{j-1} \chi_{E_j} \qquad \psi_\varepsilon = \sum_{j=1}^{n} y_j \chi_{E_j}$$

■

Lemma 10.3

sps $f : E \longrightarrow \overline{\mathbb{R}}$ LM. then, $\exists$ seq of simple $\phi_n \to f$ ptwise s.t. $|\phi_n| \leq |f|$.

if $f \geq 0$, then furthermore ϕ_n non-neg increasing.

Proof.

general follows from non-neg case: note that we have $\phi_n \to (f)_+, \psi_n \to (f)_-$ with $\phi_n \leq (f)_+, \psi_n \leq (f)_-$. thus $\phi_n - \psi_n \to (f)_+ - (f)_- = f$, and:

$$|\phi_n - \psi_n| = \phi_n + \psi_n \leq (f)_+ + (f)_- = |f|$$

for $f \geq 0$, consider $f_n(x) = \min(f(x), n)$ LM. by lemma, $\exists \phi_n$ simple s.t. $\phi_n \leq f_n$ and $|f_n(x) - \phi_n(x)| \leq 1/n$. let $\psi_1 = \max(\phi_1, 0)$, and $\psi_n = \max(\phi_n, \psi_{n-1})$. claim:

i) ψ_n are simple: by ind, ψ_n is max of two simples thus simple (fairly clear)

ii) ψ_n are increasing: clear

iii) $\psi_n \to f$: by construction + ind, $\phi_n \leq \psi_n \leq f_n \leq f$ for all n

■

$$n(n-1)2^{n-2} = \sum_{k=2}^{n} k(k-1)\binom{n}{k}$$